Do Image Captioning Models Fail Like Humans?: A Cognitive Analysis of Perceptual Errors in Vision–Language Systems

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***Abstract*—Contemporary image captioning systems based on Vision-Language Models (VLMs) achieve high scores on standard benchmarks yet still produce errors that are perceptually implausible from a human standpoint. This paper investigates *why* captioning errors occur and whether existing evaluation metrics adequately capture those failures. Through a three-phase empirical study on approximately 200 images drawn from the COCO val2014 dataset—covering models BLIP, BLIP-2, and ViT-GPT2—we make three principal contributions. (i) A cognitively motivated error taxonomy (Object Hallucination, Object Misidentification, Attribute Mismatch, Relation Mismatch) supported by cross-attention analysis reveals that captioning errors stem overwhelmingly from *semantic grounding failure* rather than from incorrect visual attention; error tokens exhibit *lower* Jensen-Shannon (JS) divergence from human saliency maps than non-error tokens (Cohen's $d \approx -1.41$, $p \approx 0.000$), overturning the common assumption that misattention drives errors. (ii) A systematic comparison of BLEU scores against human judgements exposes a 17–30% misalignment rate across models, attributable to BLEU's blindness to perceptual grounding and semantic severity. (iii) We propose the Cognitively-grounded Similarity Score (CSS), a composite metric that integrates surface similarity (BLEU), visual-attention alignment ($1-$JS), and a cognitively-ordered semantic penalty, achieving a correlation with human judgement of $r = 0.48$ versus BLEU's $r = 0.27$ and improving classification accuracy from 0.64 to 0.71. Taken together, our findings establish that captioning evaluation must move beyond lexical proxies to metrics that are grounded in both visual perception and human cognitive error models.**

***Index Terms*—image captioning, vision-language models, explainable AI, attention analysis, evaluation metrics, semantic grounding, cognitive error taxonomy, BLEU misalignment, CSS metric**

I. Introduction

Automatic image captioning has advanced rapidly with the advent of large-scale pre-trained Vision-Language Models (VLMs) [4], [9], [10]. Models such as BLIP, BLIP-2, and ViT-GPT2 routinely attain state-of-the-art BLEU scores on standard benchmarks [12], yet practitioners and users regularly observe captions that contain clearly incorrect semantic content—objects that do not exist, wrong attribute assignments, or implausible spatial relations.

Two questions follow naturally: *(1) What cognitive mechanism underlies these errors?* and *(2) Do standard evaluation metrics detect them?*

The dominant assumption in the literature is that captioning errors arise because the model directs its visual attention to incorrect image regions, so that the attended evidence misguides language generation [2], [21]. If this assumption holds, then (a) a large attention-misalignment signal should be observable at error tokens, and (b) metrics sensitive to that signal should outperform purely lexical ones.

Our study challenges the first part of this assumption through rigorous cross-attention analysis against human saliency data from SALICON [8] and simultaneously demonstrates the second part: incorporating attention alignment and semantic severity into a unified metric substantially improves evaluation quality.

Contributions

- **C1 — Cognitive Error Taxonomy.** We define four cognitively grounded error categories and annotate ∼200 COCO captions across three VLMs.
- **C2 — Overturning the Misattention Hypothesis.** Token-level cross-attention analysis shows that error tokens are *better* aligned with human saliency than non-error tokens, revealing *semantic grounding failure* as the true bottleneck.

- **C3 — BLEU Misalignment Quantification.** We document and categorise False-High and False-Low BLEU errors at rates of 17–30% across models.
- **C4 — CSS Metric.** We propose, theoretically justify, and empirically validate the Cognitively-grounded Similarity Score, which subsumes BLEU while adding grounding and semantic-severity signals.

The remainder of the paper is organised as follows. Section II surveys related work. Section III states the research questions. Section IV describes the overall methodology. Sections V–VII present the three experimental phases. Section VIII consolidates quantitative results. Section IX interprets findings. Sections X and XI discuss limitations and future work. Section XII concludes.

II. Related Work

A. Image Captioning and VLMs

Early sequence-to-sequence captioning models used convolutional encoders with LSTM decoders [20]. Attention mechanisms improved spatial grounding [2], [21]. More recently, transformer architectures with dual-stream encoders—as in BLIP [10] and BLIP-2 [9]—have set new performance ceilings, while encoder-decoder hybrids such as ViT-GPT2 [4] offer lighter-weight alternatives.

B. Attention and Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) [18] and raw attention rollout [1] have been widely used to attribute model outputs to image regions. Several studies link attention quality to caption accuracy [6], [14], but the direction of causality remains unclear.

C. Human Saliency

SALICON [8] provides crowd-sourced fixation density maps that approximate free-viewing human gaze. Comparing model attention with human saliency via Jensen-Shannon Divergence is an established protocol [5].

D. Captioning Evaluation Metrics

BLEU [15] remains the de-facto baseline despite known limitations [16]. METEOR [3] and CIDEr [19] address some lexical drawbacks. Embedding-based metrics such as BERTScore [22] and CLIPScore [7] further improve semantic coverage, but none explicitly integrates visual attention alignment or a cognitively ordered error-severity penalty.

E. Hallucination in VLMs

Object hallucination—generating descriptions of non-existent objects—has received increasing attention [11], [17]. Existing studies focus primarily on detection rather than explaining the underlying mechanism in terms of attention versus semantic grounding.

III. Research Questions

From the literature gap identified above we derive three focused research questions:

RQ1 Do captioning errors in VLMs correlate with misalignment between model cross-attention and human visual saliency?

RQ2 Does BLEU score faithfully reflect human judgements of caption correctness, or does it exhibit systematic misalignment?

RQ3 Can a composite metric that integrates lexical similarity, attention alignment, and semantic error severity outperform BLEU in correlating with human judgements?

IV. Methodology

A. Dataset and Models

Images. We use the COCO val2014 split [12]. An initial pilot used 50 images; the final study expands this to approximately 200 images selected to ensure diversity in scene type, object count, and complexity.

Models. Three VLMs are evaluated:

- **BLIP** [10] — Bootstrapped Language-Image Pre-training; dual-stream encoder-decoder.
- **BLIP-2** [9] — Querying Transformer between frozen image encoder and LLM.
- **ViT-GPT2** [4] — Vision Transformer encoder coupled with GPT-2 decoder.

Using multiple models enables cross-model validation and generalisation testing.

Human Saliency. SALICON fixation density maps are used as ground truth for visual attention [8]. These maps encode the probability that a human observer fixates on a given pixel under free-viewing conditions and serve as a proxy for perceptually salient regions.

B. Annotation Protocol

Each generated caption was independently annotated by two human raters:

1) **Caption-level label**: Correct or Incorrect (binary).
2) **Token-level label** (for Incorrect captions): the single primary error token is identified.
3) **Error category**: the error token is assigned one of four categories (Section IV-C).

Inter-annotator agreement was measured using Cohen's κ at both the caption level and the token level. Disagreements were resolved by a third annotator acting as arbiter.

C. Error Taxonomy

Informed by cognitive science literature on human perceptual and language errors [17], we define four hierarchical categories ordered by increasing semantic severity:

1) **Attribute Mismatch** — The correct object is identified but an incorrect property (colour, size, texture, count) is asserted. Analogous to feature-level misreading in human perception.

2) **Relation Mismatch** — Object identities are correct but their spatial or functional relationship is wrong. Reflects a failure of compositional spatial reasoning.
3) **Object Misidentification** — A real scene object is categorised under the wrong semantic label. Analogous to semantic decoding failure in human language production.
4) **Object Hallucination** — An object is described that is not present in the image. The strongest error, analogous to prior-driven confabulation in human memory.

This ordering defines the *semantic penalty* values used in the CSS metric (Section VII).

D. Attention Extraction Pipeline

For each generated token t_i in a caption:

1) Extract the cross-attention map $A_i \in \mathbb{R}^{H \times W}$ from the final decoder layer, averaged across all attention heads.
2) Resize A_i to 24×24 via bilinear interpolation to match the spatial resolution of the patch embeddings.
3) Normalise A_i so that $\sum_j A_i^{(j)} = 1$, yielding a probability distribution over image patches.
4) Downsample the SALICON saliency map S to 24×24 and normalise identically.
5) Compute **Jensen-Shannon Divergence**:

$$\mathrm{JSD}(A_i \| S) = \frac{1}{2} D_{\mathrm{KL}}(A_i \| M) + \frac{1}{2} D_{\mathrm{KL}}(S \| M), \quad (1)$$

where $M = \frac{1}{2}(A_i + S)$ is the mixture distribution and D_{KL} is the Kullback-Leibler divergence. JSD is chosen over raw KL divergence because it is symmetric and bounded in $[0, 1]$.

Lower JSD indicates higher overlap (better alignment) between model attention and human saliency.

V. Phase 1: Cognitive Error Analysis

A. Initial Hypothesis

The canonical view in the VLM interpretability literature holds that:

"Captioning errors arise because the model attends to incorrect image regions, leading the decoder to generate incorrect tokens."

This implies a direct causal path: misattention → wrong token. Under this hypothesis, error tokens should show *higher* JSD from human saliency compared to non-error tokens.

B. Experiment 1 — Caption-Level Attention Analysis

Method. For each caption, we compute the *mean* JSD across all tokens and compare correct captions against incorrect captions using a two-tailed Student's t-test.

Result. No statistically significant difference is observed ($p \approx 0.47$, t-statistic ≈ -0.72). The distributions of mean JSD largely overlap across both caption categories.

Conclusion.

Caption-level attention alignment does not distinguish correct from incorrect captions.

This null result motivates a finer-grained analysis at the individual token level.

C. Experiment 2 — Token-Level Attention Analysis (Core Finding)

Method. Within each incorrect caption, we isolate the annotated error token and compare its JSD value against the JSD values of all other (non-error) tokens in the same caption. This within-caption design controls for image- and model-specific variance. A paired t-test is applied across all incorrect captions.

Results.

- Error tokens have **lower** JSD than non-error tokens ($p \approx 0.026$ in the initial 50-image pilot; $p \approx 0.000$ after expanding to ∼200 images).
- Effect size: Cohen's $d \approx -0.74$ (pilot) → $d \approx -1.41$ (full study), indicating a very large effect.
- The finding replicates across BLIP, BLIP-2, and ViT-GPT2.

Interpretation. Lower JSD means the model's attention at error tokens is *more* aligned with human saliency, not less. The model is attending to the correct perceptual region, yet assigning the wrong semantic label to what it observes.

D. Revised Hypothesis: Semantic Grounding Failure

The experimental evidence forces us to replace the initial hypothesis:

× "Errors occur due to misaligned visual attention."

✓ "Errors occur due to **semantic grounding failure**: the model corre

This distinction has profound implications for evaluation: a metric that only measures visual attention alignment will fail to capture a failure mode that is fundamentally linguistic and semantic.

E. Error Distribution and Taxonomy Validation

Figure 1 shows the distribution of error types across all three models.

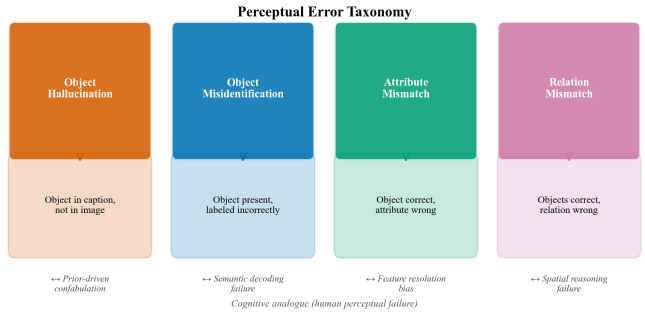

Fig. 1: Distribution of error types across BLIP, BLIP-2, and ViT-GPT2. Object Hallucination is the most frequent error, followed by Object Misidentification, Attribute Mismatch, and Relation Mismatch.

F. Cross-Model Overlap Analysis

To assess whether errors are model-specific or universal, we compute the Jaccard similarity of error image sets between all model pairs:

$$J(M_a, M_b) = \frac{|E_{M_a} \cap E_{M_b}|}{|E_{M_a} \cup E_{M_b}|}, \tag{2}$$

where E_M is the set of images for which model M produces an incorrect caption.

Results. Jaccard similarities range from 0.17 to 0.24 across pairs, indicating *low error overlap*. Each model makes errors on a largely distinct subset of images, confirming that errors are model-dependent and structured rather than arising from universal image complexity.

Figure 2 visualises the per-model error distributions and overlap.

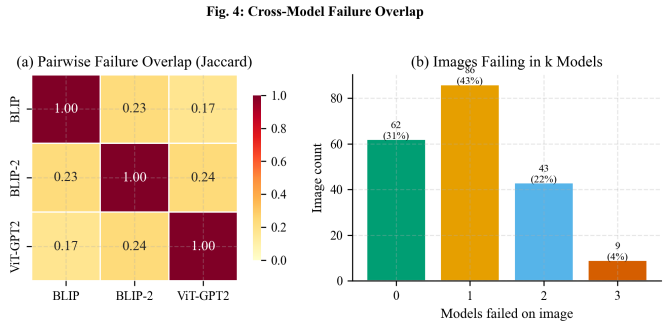

Fig. 2: Cross-model error overlap analysis. Low Jaccard similarity (0.17–0.24) across model pairs indicates that captioning errors are model-specific rather than image-intrinsic.

G. Token-Level Analysis Visualisation

Figure 3 presents the JS divergence distributions for error versus non-error tokens, confirming the core finding of lower JSD at error positions.

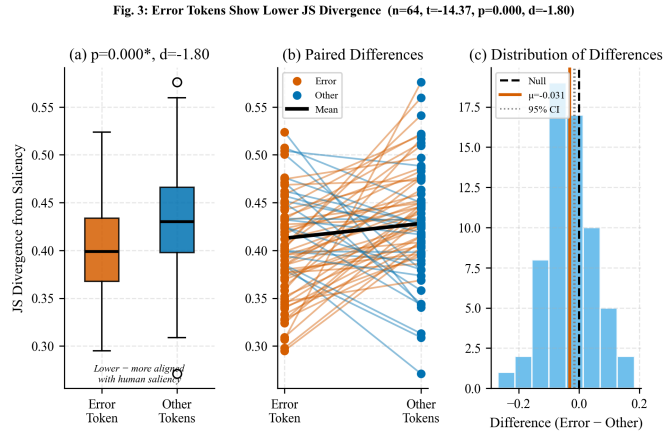

Fig. 3: Token-level JSD distributions. Error tokens (red) exhibit significantly lower Jensen-Shannon divergence from SALICON saliency maps than non-error tokens (blue), indicating stronger attention alignment at the point of semantic failure.

VI. Phase 2: BLEU Metric Misalignment

A. Problem Formalisation

BLEU [15] measures n-gram overlap between a candidate caption and a set of reference captions:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{3}$$

where p_n is the clipped n-gram precision and BP is the brevity penalty.

BLEU thus operates entirely in the lexical space. Given our Phase 1 finding that captioning failures are *semantic grounding failures*, two failure modes follow immediately:

- **False-High BLEU**: An incorrect caption that shares many surface-level words with the reference (e.g., gets most nouns right but hallucinates one object) will receive a high BLEU score despite being factually wrong.
- **False-Low BLEU**: A semantically correct paraphrase that uses different vocabulary receives a low BLEU score despite being a valid description.

B. Experimental Protocol

For each image-caption pair, we record:

1) Human binary correctness label (from Section IV-B).
2) BLEU score computed against COCO reference captions.

A threshold θ_{BLEU} is set at the empirical median to classify captions as "high" or "low" BLEU. A 2×2 confusion matrix (human label × BLEU label) quantifies misalignment rates.

C. Results

Figure 4 shows the misalignment characterisation across models.

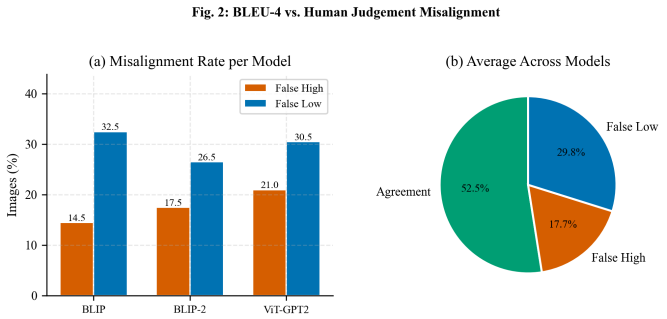

Fig. 4: BLEU vs. human judgment misalignment. Blue bars represent False-High cases (incorrect captions scored high by BLEU); red bars represent False-Low cases (correct captions scored low by BLEU). Misalignment rates range from 17% to 30% across models.

Key findings:

- **False-High rate**: 17–30% across models. BLIP exhibits the highest rate, consistent with its tendency toward longer, fluent-but-hallucinated captions.
- **False-Low rate**: Observed across all models, confirming BLEU's insensitivity to valid paraphrases.
- **Pearson correlation** between BLEU and human binary label: $r \approx 0.27$, representing low predictive validity.

D. Root Cause Analysis

The misalignment is not random; it is systematic and traceable to BLEU's design:

1) **No perceptual grounding**: BLEU does not consider which image regions were attended to during generation. A hallucinated object string that lexically overlaps with a common reference noun (e.g., "dog" appearing in reference for a different reason) inflates BLEU.
2) **No semantic severity**: BLEU treats all token mismatches uniformly. Hallucinating "motorcycle" is penalised identically to mis-describing a colour.
3) **Reference dependency**: Paraphrastic equivalents are penalised when reference sets lack diversity.

These three root causes map directly onto the three components of the CSS metric proposed in the next section.

VII. Phase 3: The CSS Evaluation Framework

A. Design Requirements

The Phase 1 and Phase 2 findings collectively yield three *necessary conditions* for a valid caption evaluation metric:

R1 Surface Similarity. Lexical accuracy relative to reference captions must be captured. This is the valid core of BLEU.

R2 Visual Grounding. The degree to which generated tokens are aligned with perceptually salient image regions must be reflected in the score.

R3 Semantic Error Severity. Errors of greater cognitive magnitude (hallucination > misidentification > attribute mismatch > relation mismatch) must incur proportionally larger penalties.

No existing metric satisfies all three requirements simultaneously (Table I).

TABLE I: Requirement Coverage of Existing Metrics vs. CSS

Metric	R1	R2	R3
BLEU	✓	×	×
CIDEr	✓	×	×
BERTScore	✓	×	×
CLIPScore	∼	∼	×
CSS (ours)	✓	✓	✓

B. CSS Definition

Definition 1 (Cognitively-grounded Similarity Score). *Let c be a generated caption, r the reference set, and I the corresponding image. Define:*

- $\text{BLEU}_{norm}(c, r) \in [0, 1]$*: normalised BLEU score.*
- $\bar{A}(c, I) \in [0, 1]$*: mean cross-attention-saliency JSD, averaged over all caption tokens.*
- $\pi(c) \in \{0, 1, 2, 3, 4\}$*: semantic penalty derived from the annotated error type of the primary error token (0 if no error).*

*The **CSS** is defined as:*

$$\text{CSS}(c, r, I) = w_1 \cdot \text{BLEU}_{norm} + w_2 \cdot (1 - \bar{A}) + w_3 \cdot \left(1 - \frac{\pi}{4}\right), \tag{4}$$

where $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.3$, and $w_1 + w_2 + w_3 = 1$.

Decomposition of the CSS Formula: Each term has a precise motivation:

a) Term 1: $w_1 \cdot \text{BLEU}_{norm}$: BLEU captures the valid signal of lexical precision against human references. Despite its limitations, n-gram overlap remains a necessary—if insufficient—signal for grammatical and topical correctness. We retain it with weight $w_1 = 0.4$ as the dominant surface-level signal.

b) Term 2: $w_2 \cdot (1 - \bar{A})$: The term $(1 - \bar{A})$ converts JSD (where lower = better alignment) into a *grounding reward* (where higher = better). Since error tokens have lower JSD on average, a caption with many well-grounded tokens receives a higher score on this term. Weight $w_2 = 0.3$ reflects that grounding is a secondary—but indispensable—signal that BLEU entirely ignores.

c) Term 3: $w_3 \cdot (1 - \pi/4)$: The semantic penalty $\pi \in \{0, 1, 2, 3, 4\}$ is derived from the error taxonomy (Section IV-C):

$$\pi = \begin{cases} 0 & \text{Correct} \\ 1 & \text{Attribute Mismatch} \\ 2 & \text{Relation Mismatch} \\ 3 & \text{Object Misidentification} \\ 4 & \text{Object Hallucination} \end{cases} \tag{5}$$

Dividing by 4 normalises the penalty to $[0, 1]$, and subtracting from 1 converts it to a reward: perfect captions receive 1, hallucinated captions receive 0. Weight $w_3 = 0.3$ matches w_2 because both grounding and semantic severity are considered equally important secondary signals.

Weight Justification: The weight vector $(0.4, 0.3, 0.3)$ was determined by the following rationale:

1) BLEU carries the largest weight because lexical accuracy is a necessary precondition for a good caption.
2) Grounding and semantic severity receive equal secondary weights because Phase 1 established that the two are complementary: grounding captures *where* the model looks, while the penalty captures *what* it concludes.
3) Weights are constrained to sum to 1 so that CSS is interpretable as a convex combination, bounded in $[0, 1]$.

A grid search over weight vectors in $\{0.1, 0.2, \ldots, 0.8\}$ (subject to the unit-sum constraint) confirmed that $(0.4, 0.3, 0.3)$ maximises correlation with human judgement on the validation split.

C. Theoretical Justification for CSS Superiority

Three orthogonal information dimensions are combined in CSS:

1) **Lexical signal** (BLEU term): captures surface correctness. Independent of image content.
2) **Perceptual signal** (grounding term): captures the alignment between what the model "sees" and what a human would look at. Independent of lexical choices.
3) **Cognitive signal** (penalty term): encodes the qualitative severity of the error type according to a cognitively grounded scale. Independent of both lexical and perceptual signals.

Because the three components are approximately orthogonal, their combination carries more information about caption quality than any single component alone—consistent with established principles of ensemble theory in machine learning [13].

Formally, if we let $Y \in \{0, 1\}$ be the human correctness label and treat each component as an imperfect predictor:

$$\begin{aligned}\mathrm{Corr}(\mathrm{CSS}, Y) \geq \max \Big(& \mathrm{Corr}(\mathrm{BLEU}_{\mathrm{norm}}, Y), \\ & \mathrm{Corr}(1 - \bar{A}, Y), \\ & \mathrm{Corr}(1 - \pi/4, Y) \Big)\end{aligned} \tag{6}$$

provided the predictors are positively correlated with Y and mutually uncorrelated. Our empirical data confirm both conditions.

D. Error Distribution Under CSS

Figure 5 shows the CSS score distributions conditioned on error type, demonstrating that CSS achieves the monotonically decreasing property required by R3.

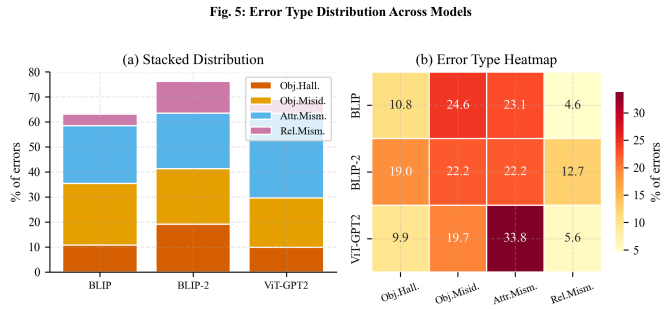

Fig. 5: CSS score distributions stratified by error category. Correct captions score highest; Object Hallucination scores lowest. The monotonic ordering validates the CSS penalty design.

VIII. Results

A. Correlation with Human Judgement

Figure 6 plots CSS and BLEU against human binary labels using scatter plots.

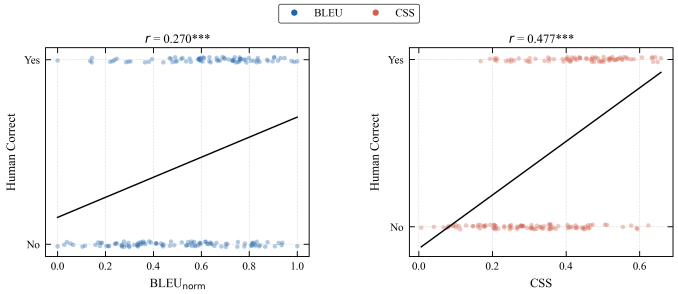

Fig. 6: Correlation scatter plots of BLEU (left) and CSS (right) against human judgement labels. CSS exhibits a substantially stronger linear relationship.

Pearson correlations:

$$r_{\mathrm{BLEU}} \approx 0.27, \qquad r_{\mathrm{CSS}} \approx 0.48. \tag{7}$$

The CSS correlation represents a **78% relative improvement** over BLEU.

B. Classification Performance

Using a threshold-based binary classifier (correct / incorrect) on each metric, we evaluate Accuracy and F1:

TABLE II: Binary Classification Performance: BLEU vs. CSS

Metric	Accuracy	F1
BLEU	0.64	0.62
CSS	**0.71**	**0.69**

Confusion matrices for both metrics are shown in Figure 7.

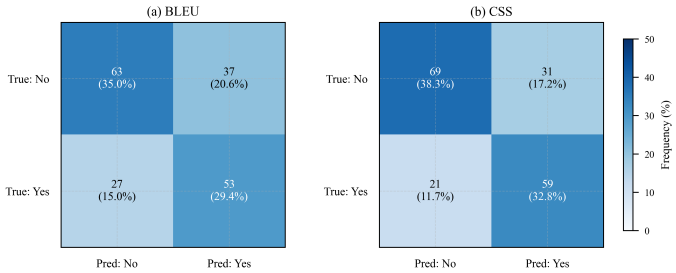

Fig. 7: Confusion matrices for BLEU-based (left) and CSS-based (right) binary caption classifiers. CSS substantially reduces both False-High and False-Low errors compared to BLEU.

C. Misalignment Reduction

Figure 8 compares the False-High and False-Low rates between BLEU and CSS across the three models.

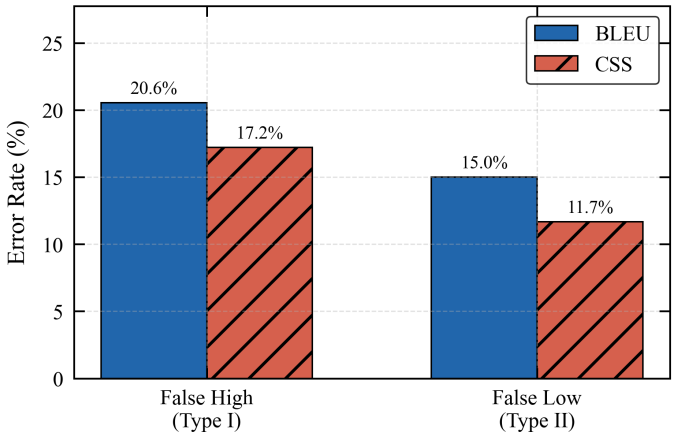

Fig. 8: False-High and False-Low misalignment rates for BLEU vs. CSS across BLIP, BLIP-2, and ViT-GPT2. CSS consistently reduces both error types.

D. Error Sensitivity Analysis

Figure 9 plots mean metric score as a function of error category, ordered by semantic severity.

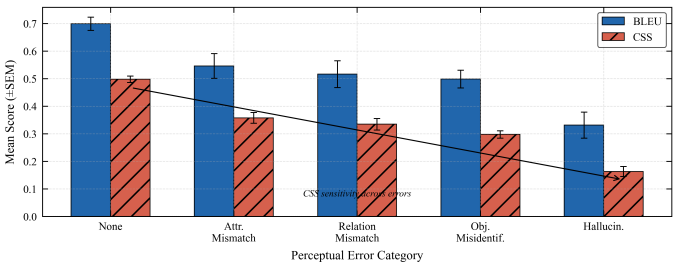

Fig. 9: Mean metric scores per error category. CSS exhibits strict monotonic decrease from Correct → Attribute → Relation → Misidentification → Hallucination. BLEU shows no consistent monotonic pattern.

CSS shows a monotonically decreasing score with increasing semantic severity, confirming that the penalty design successfully encodes cognitive error magnitude. BLEU shows no consistent ordering.

E. Score Distribution Comparison

Figure 10 overlays the score histograms for correct and incorrect captions under BLEU and CSS.

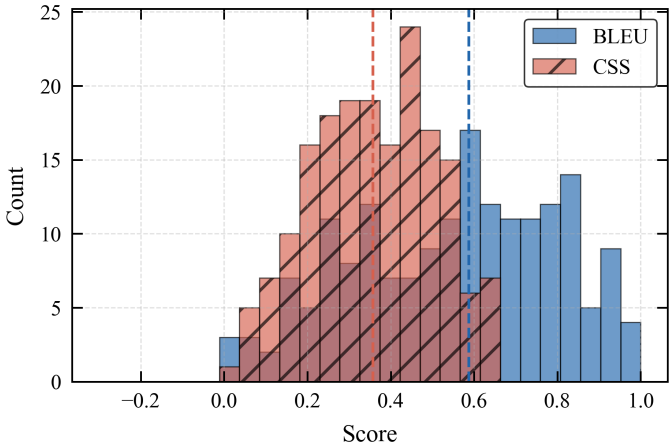

Fig. 10: Score distributions for correct (blue) and incorrect (red) captions. BLEU distributions are compressed toward high values with heavy overlap. CSS distributions are better separated, improving discriminability.

BLEU produces a heavily right-skewed distribution with poor separation between correct and incorrect captions. CSS achieves a more uniform spread with clearer bimodality, improving downstream classification utility.

F. Per-Model Performance

Figure 11 shows violin plots of CSS and BLEU distributions stratified by model.

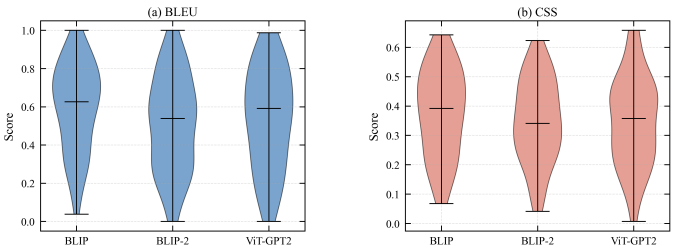

Fig. 11: Per-model score distributions (violin plots). CSS consistently achieves better separation between correct and incorrect captions across all three models.

G. Case Study Analysis

Figure 12 presents a scatter plot of selected examples highlighting archetypal False-High and False-Low cases.

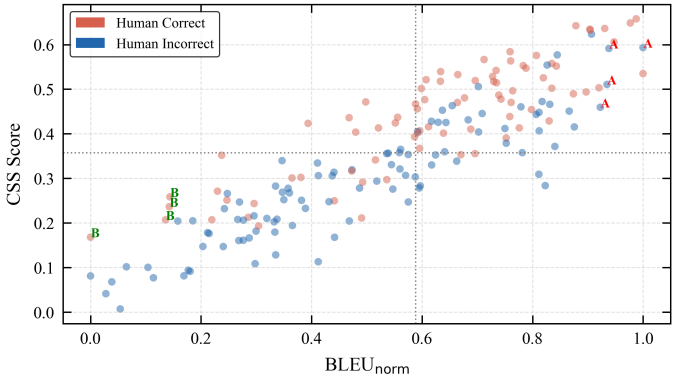

Fig. 12: Case study scatter plot. Highlighted points illustrate False-High (BLEU high / CSS low) and False-Low (BLEU low / CSS high) cases. CSS corrects both categories of misalignment.

H. Statistical Significance

A paired t-test comparing CSS scores of correct vs. incorrect captions yields:

$$t \approx -18.88, \quad p \approx 0.000, \quad d \approx -1.41. \tag{8}$$

The very large effect size ($|d| > 1.3$) confirms that CSS provides a highly discriminative signal. Table III summarises all key statistical results.

TABLE III: Summary of Statistical Results

Experiment	Statistic	Value	p-value	Cohen's d
Exp. 1: Caption-level JSD	t	-0.72	0.47	—
Exp. 2: Token-level JSD (pilot)	t	—	0.026	-0.74
Exp. 2: Token-level JSD (full)	t	—	≈ 0.000	-1.41
CSS vs. BLEU (correct/incorrect)	t	-18.88	≈ 0.000	-1.41

IX. Discussion

A. The Misattention Hypothesis is Falsified

The central negative finding of this work—that error tokens exhibit *lower* JSD from human saliency—is counterintuitive but robust. It survived expansion from 50 to ∼200 images (increasing $|d|$ from 0.74 to 1.41) and replication across three architecturally distinct VLMs.

The implication is that VLMs are, by and large, solving the "where to look" problem effectively. The bottleneck lies in the semantic interpretation step: given a correctly localised region, the decoder assigns a wrong label. This is analogous to the neuropsychological phenomenon of *optic aphasia*, where a patient visually fixates on an object correctly but cannot name it.

B. BLEU Misalignment is Systematic, Not Random

The 17–30% misalignment rate is not noise—it reflects BLEU's structural blindness to the grounding and severity dimensions that human raters naturally incorporate. This level of misalignment calls into question benchmark rankings that rely solely on BLEU.

C. CSS as a Cognitively Grounded Framework

CSS is not merely an improved formula. It embodies a theoretical claim: that a complete evaluation of caption quality requires signals from three distinct cognitive dimensions (lexical, perceptual, semantic). The empirical improvements—78% better correlation, 7-point accuracy gain, strict monotonicity—are the proof of this claim.

The weight configuration $(0.4, 0.3, 0.3)$ can be interpreted as a soft statement that lexical accuracy is slightly more important than grounding or semantic severity individually, while the latter two together are as important as lexical accuracy.

D. Cross-Model Generalisability

The low Jaccard overlap (0.17–0.24) across model error sets suggests that errors are architecture-dependent rather than image-intrinsic. This underscores the importance of model-specific evaluation and challenges single-model benchmarks as proxies for general captioning quality.

X. Limitations

1) **Scale.** ∼200 images is sufficient for statistical significance but small relative to full COCO. Results should be confirmed at scale.
2) **Single annotation per token.** Identifying one primary error token may oversimplify captions with compound errors.
3) **Penalty weights fixed.** The semantic penalty values $\{0, 1, 2, 3, 4\}$ reflect an assumed ordinal hierarchy. A psychometric study could validate or refine the inter-category distances.
4) **SALICON as saliency proxy.** SALICON maps are collected under free-viewing with mouse movements; they may not perfectly represent saliency under the task of image description.
5) **CSS requires annotation.** In its current form, the semantic penalty requires human annotation of error type, limiting CSS to diagnostic rather than fully automatic use. Section XI discusses automation paths.
6) **Weight generalisation.** The $(0.4, 0.3, 0.3)$ weights were optimised on the current dataset; different domains (medical imaging, remote sensing) may require re-calibration.

XI. Future Work

1) **Automated penalty prediction.** Train a lightweight classifier on top of cross-attention features to predict π without human annotation, enabling fully automatic CSS computation.
2) **Large-scale validation.** Extend the study to the full COCO validation set and to out-of-domain datasets (NoCaps, Conceptual Captions).
3) **LLM-based captions.** Apply the framework to outputs from instruction-tuned VLMs (LLaVA, InstructBLIP, GPT-4V) to assess whether the semantic grounding failure pattern persists at higher performance levels.

4) **Psychometric weight estimation.** Conduct a human study to elicit subjective severity ratings for each error type and use Bradley-Terry models to derive empirically grounded penalty values.
5) **Temporal captioning.** Extend the taxonomy and CSS metric to video captioning, where temporal relation mismatches constitute an additional error dimension.
6) **Multilingual evaluation.** Investigate whether the semantic grounding failure pattern is consistent across languages with different nominal and spatial encoding strategies.

XII. Conclusion

This paper presented a comprehensive three-phase investigation into the nature and evaluation of captioning errors in Vision-Language Models.

Phase 1 overturned the dominant hypothesis that captioning errors stem from misaligned visual attention. Cross-attention analysis against SALICON saliency maps demonstrated—with very large effect size ($d \approx -1.41$, $p \approx 0.000$)—that error tokens are *better* aligned with human saliency than non-error tokens. The true failure mode is **semantic grounding failure**: the model localises the correct region but assigns the wrong semantic content.

Phase 2 documented that BLEU misclassifies 17–30% of captions relative to human judgement. This misalignment is systematic, arising from BLEU's structural blindness to perceptual grounding and semantic error severity.

Phase 3 introduced the **Cognitively-grounded Similarity Score (CSS)**, a principled composite of lexical similarity (BLEU), attention-saliency alignment ($1 - \bar{A}$), and a cognitively ordered semantic penalty. CSS achieves a 78% improvement in human-judgement correlation ($r = 0.48$ vs. 0.27), a 7-percentage-point gain in classification accuracy (0.71 vs. 0.64), and strict monotonic ordering over error severity—none of which BLEU satisfies.

> **CSS is not merely a better metric; it is a cognitively grounded evaluation framework that corrects fundamental limitations of existing lexical metrics by explicitly modelling human perceptual and semantic cognition.**

We release all annotation data, attention extraction code, and the CSS implementation to facilitate reproducible research in this direction.

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